

Manually Calculating SRS & SOS

Companion Piece with “Predicting March Madness”

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Motivation

Sports-Reference has two statistics available for every professional team and major program: **Simple Rating System** (*SRS*) and **Strength of Schedule** (*SOS*). *SRS* and *SOS* were created by Doug Drinen as a ranking algorithm for the 2005 NFL season. *SOS* evaluates how difficult a team’s schedule is relative to their peers and *SRS* is the sum of a team’s average point differential with their *SOS*. It’s particularly meaningful in the NFL where 32 teams play 16 games and six games against divisional opponents, so certain teams can have relatively softer schedules if their division is weaker. Another place this applies: College Basketball.

In my March Madness machine learning model, I included the teams’ *SRS* and *SOS* as predictors. However, the values on *Sports-Reference* account for the March Madness games, so there would be data leakage. I can’t use values to predict March Madness outcomes if those values already reflect the results; it’s cheating and it’s counterproductive, because the model would be poorly built for new data.

Hello Again, Linear Algebra

Using Drinen’s article, my goal was to craft the same system of equations. A team rating is calculated by adding their average point differential with the mean of their opponents’ team ratings. In the NFL example, there are 32 team ratings, meaning that there are 32 unknown variables and equations to solve. Unfortunately, in the case of

$$\begin{aligned} R_{\text{ind}} &= 12.0 + (1/16) (R_{\text{bal}} + R_{\text{jax}} + R_{\text{cle}} + \dots + R_{\text{ari}}) \\ R_{\text{pit}} &= 8.2 + (1/16) (R_{\text{ten}} + R_{\text{hou}} + R_{\text{nwe}} + \dots + R_{\text{det}}) \\ &\cdot \\ &\cdot \\ &\cdot \\ R_{\text{stl}} &= -4.1 + (1/16) (R_{\text{sfo}} + R_{\text{ari}} + R_{\text{ten}} + \dots + R_{\text{dal}}) \end{aligned}$$

Figure 1: NFL team rating calculation 2005 example

the NCAA, there are more than 350 teams. I needed to calculate the team rating for all the teams, not just the 64 tournament teams, since solving for the rating of the March teams requires knowing every team's rating.

Process

There is a caveat in the case of college basketball. One, teams often have fixtures against anywhere from one to three non-Division-I opponents. Those games are completely excluded from the calculation. They do not factor into the calculation for point differential and they're not included in the system of equations.

Using BeautifulSoup, I navigated to *Sports-Reference* and scraped each school's schedule prior to any March Madness or NIT¹ games from 2010-2025. I built a CSV to get a record of their opponents and their average point differential (Figure 1).

team	MoV	opp_1	opp_2	opp_3
Abilene Christian	-5.933333333333334	Omaha	Stephen F. Austin	Texas State
Air Force	-18.34375	Belmont	Austin Peay	Long Island University
Akron	12.375	James Madison	Princeton	Purdue
Alabama	8.25	North Dakota	St. John's (NY)	Purdue
Alabama A&M	-2.5	Indiana	Charleston Southern	Lindenwood
Alabama State	-6.064516129032258	Florida State	UAB	Colorado
Albany (NY)	-5.827586206896552	Marquette	Massachusetts	Rhode Island
Alcorn State	-13.28125	Florida State	South Alabama	Minnesota
American	-0.7	Wake Forest	Pennsylvania	George Washington
Appalachian State	1.9	Central Michigan	North Carolina Central	Ohio State
Arizona	17.352941176470587	Florida	Utah Tech	Northern Arizona
Arizona State	-2.0	Southern Utah	Utah Tech	Gonzaga

Figure 2: 2026 Schedule Dataset Example

Treating each school name as a variable, we can build the system of equations.

$$R_{school} = MoV_{school} + \frac{1}{n} (R_{opp1} + R_{opp2} + \dots + R_{oppn}) \quad (1)$$

$$-MoV_{school} = -R_{school} + \frac{1}{n} (R_{opp-1} + R_{opp-2} + \dots + R_{opp-n}) \quad (2)$$

Where:

- n = The number of games a team played against D-I opponents
- R_{school} = A team's rating, currently unknown
- MoV = Average Margin of Victory

$$MoV = \frac{1}{n} \sum_{i=1}^n PS_i - PA_i \quad (3)$$

Where:

- PS = points scored
- PA = points allowed

¹Or CBC or CBI or CIT, am I missing any more?

- Each i is a game played

The article demonstrates how to derive SOS and SRS from these team ratings:

$$SOS = \frac{1}{n} (R_{opp.1} + R_{opp.2} + \dots + R_{opp.n}) \quad (4)$$

$$SRS = MoV + SOS \quad (5)$$

Problem Solved

Using this pipeline and algorithm, I can now include two meaningful predictors without having to worry about data leakage. Having SRS or SOS included that take March Madness games into account would greatly affect the model's ability to predict. It would essentially be "overfitting" because it was given a hint, which would provide a huge boost in training accuracy. This issue would not even be discernible using a test/valid set or cross validation, because all available data would have this data leakage issue. The model would struggle with any organic or new March Madness tournament field, since that hint from SRS wouldn't be present.

References

- [1] Drinen, Doug, “A very simple ranking system,” *Pro-Football Reference*, 8 May 2006, [URL](#).
- [2] Lynch, Mike, “SRS Calculation Details,” *Sports-Reference*, 3 Mar 2015, [URL](#).
- [3] “Glossary,” *Sports-Reference*, [URL](#).